
%sheikh wasimuddin CSEIOT 61 LAB 3

```
clc;
clear;
close all;
sensor.position=[40 40];
sensor.energy=120;
sensor.senseRange=15;
sensor.commRange=55;
gateway.position=[75 70];
limit=28;
txEnergy=8;
fprintf('=====\\n');
fprintf('          SINGLE NODE WSN MODEL\\n');
fprintf('=====\\n');
fprintf('\\n--- NODE INFORMATION ---\\n');
fprintf('Sensor Position          : (%.2f, %.2f)\\n', sensor.position(1), sensor.position(2));
fprintf('Initial Energy           : %.2f J\\n', sensor.energy);
fprintf('Sensing Range            : %.2f m\\n', sensor.senseRange);
fprintf('Communication Range      : %.2f m\\n', sensor.commRange);
fprintf('Base Station Position    : (%.2f, %.2f)\\n', gateway.position(1), gateway.position(2));
fprintf('\\n=====\\n');
fprintf('          SENSING PHASE\\n');
fprintf('=====\\n');
measuredValue=24+2*randn;
fprintf('Sensor is collecting environmental data...\\n');
fprintf('Measured Temperature      : %.2f degree C\\n', measuredValue);
fprintf('\\n=====\\n');
fprintf('          PROCESSING PHASE\\n');
fprintf('=====\\n');
fprintf('Temperature Limit        : %.2f degree C\\n', limit);
if measuredValue>=limit
    result=1;
    condition='HIGH';
    fprintf('Processed Result          : Temperature HIGH\\n');
else
    result=0;
    condition='NORMAL';
    fprintf('Processed Result          : Temperature NORMAL\\n');
end
fprintf('Processed Data           : %d\\n', result);
fprintf('\\n=====\\n');
fprintf('          COMMUNICATION PHASE\\n');
fprintf('=====\\n');
dx=sensor.position(1)-gateway.position(1);
dy=sensor.position(2)-gateway.position(2);
linkDistance=hypot(dx, dy);
fprintf('Distance to Base Station : %.2f m\\n', linkDistance);
if linkDistance<=sensor.commRange
    fprintf('Communication Status      : WITHIN RANGE\\n');
    fprintf('Data Transmission         : SUCCESSFUL\\n');
```

```

        sensor.energy=sensor.energy-txEnergy;
        fprintf('Transmission Energy Used    : %.2f J\n',txEnergy);
    else
        fprintf('Communication Status        : OUT OF RANGE\n');
        fprintf('Data Transmission              : FAILED\n');
    end
    fprintf('\n===== \n');
    fprintf('                ENERGY STATUS\n');
    fprintf('===== \n');
    fprintf('Initial Energy          : 120.00 J\n');
    fprintf('Remaining Energy        : %.2f J\n',sensor.energy);
    fprintf('Energy Consumed         : %.2f J\n',120-sensor.energy);
    fprintf('\n===== \n');
    fprintf('                FINAL RESULT\n');
    fprintf('===== \n');
    fprintf('Measured Temperature    : %.2f degree C\n',measuredValue);
    fprintf('Temperature Status      : %s\n',condition);
    fprintf('Processed Data          : %d\n',result);
    fprintf('Node-to-BS Distance     : %.2f m\n',linkDistance);
    if linkDistance<=sensor.commRange
        fprintf('Communication          : SUCCESSFUL\n');
    else
        fprintf('Communication          : FAILED\n');
    end
    fprintf('Remaining Node Energy    : %.2f J\n',sensor.energy);
    figure('Name','Single Node Network','Color','w');
    scatter(sensor.position(1),sensor.position(2),140,'filled','b');
    hold on;
    scatter(gateway.position(1),gateway.position(2),140,'filled','r','s');
    phi=0:0.01:2*pi;
    plot(sensor.position(1)+sensor.senseRange*cos(phi),sensor.position(2)+sensor.senseRange*sin(phi),'c-.','LineWidth',1.5);
    plot(sensor.position(1)+sensor.commRange*cos(phi),sensor.position(2)+sensor.commRange*sin(phi),'b:','LineWidth',1.8);
    if linkDistance<=sensor.commRange
        plot([sensor.position(1) gateway.position(1)],[sensor.position(2) gateway.position(2)],'g-','LineWidth',2);
    end
    text(sensor.position(1)+2,sensor.position(2)+2,'Sensor Node','FontWeight','bold');
    text(gateway.position(1)+2,gateway.position(2)+2,'Base Station','FontWeight','bold');
    xlabel('X Position (m)');
    ylabel('Y Position (m)');
    title('Single Node Wireless Sensor Network');
    grid on;
    axis equal;
    xlim([0 110]);
    ylim([0 110]);
    legend('Sensor Node','Base Station','Sensing Range','Communication Range','Communication Link','Location','best');
    hold off;

```

```

=====
SINGLE NODE WSN MODEL
=====

--- NODE INFORMATION ---
Sensor Position      : (40.00, 40.00) m
Initial Energy       : 120.00 J
Sensing Range        : 15.00 m
Communication Range   : 55.00 m
Base Station Position : (75.00, 70.00) m

=====
SENSING PHASE
=====
Sensor is collecting environmental data...
Measured Temperature  : 24.62 degree C

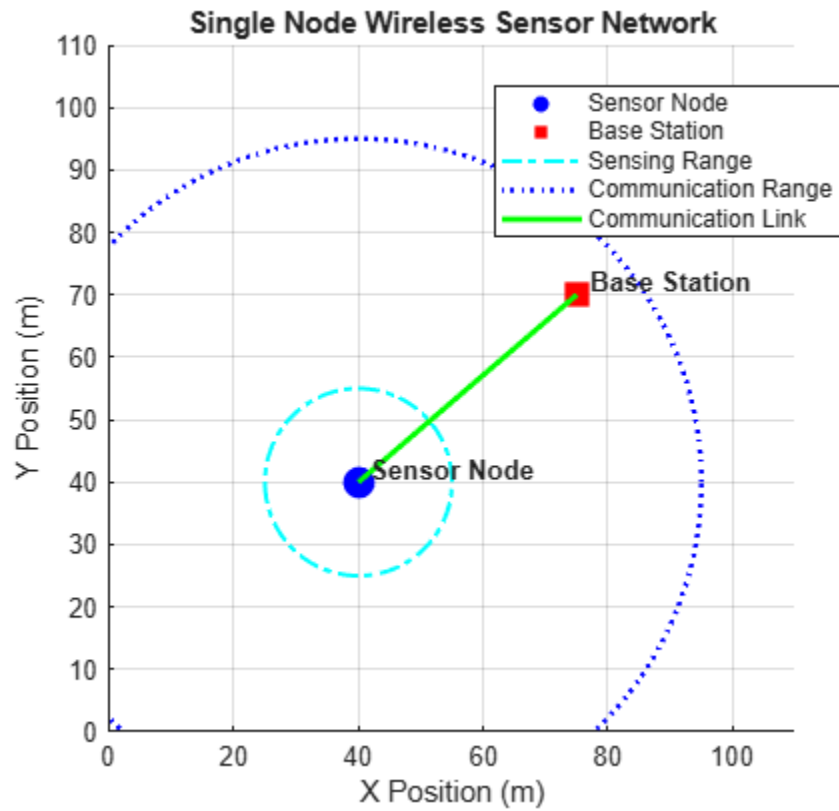
=====
PROCESSING PHASE
=====
Temperature Limit     : 28.00 degree C
Processed Result       : Temperature NORMAL
Processed Data         : 0

=====
COMMUNICATION PHASE
=====
Distance to Base Station : 46.10 m
Communication Status      : WITHIN RANGE
Data Transmission        : SUCCESSFUL
Transmission Energy Used  : 8.00 J

=====
ENERGY STATUS
=====
Initial Energy         : 120.00 J
Remaining Energy        : 112.00 J
Energy Consumed         : 8.00 J

=====
FINAL RESULT
=====
Measured Temperature    : 24.62 degree C
Temperature Status      : NORMAL
Processed Data          : 0
Node-to-BS Distance     : 46.10 m
Communication           : SUCCESSFUL
Remaining Node Energy    : 112.00 J

```



Published with MATLAB® R2026a